Change vs. uniform DoP 2 at the same rate: DoP 40 at L = 100k, M = 500k, H = 1M events/s; DS2 and Dhalion (Dh) at 1M events/s; m510

		Flink					Storm					Spark					Kafka S.					RisingW.				
		L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh
CPU	Sentiment Analysis (SA)	18	16	16	13	0.98	n/a	n/a	n/a	n/a	n/a	0.98	2.3	1.7	1.6	2.4	1	1	0.98	n/a	n/a	1	1	0.9	1	1
	Traffic Monitoring (TM)	0.57	4.2	3.3	67	64	0.63	0.61	0.59	0.081	1.3	5.1	14	22	3.7	14	0.32	0.024	0.000	n/a	779	0.98	0.57	0.98	1	1
	Trending Topics (TT)	0.72	1.5	3	66	9.2	11	110	194	3.8	0.45	0.93	1.5	1.6	0.002	0.28	0.84	1.4	1.3	0.25	0.12	0.93	1.5	3.2	3.7	6.2
S+N State	Machine Outlier (MO)	1.2	1.1	1.6	1.8	3.9	7.7	2.9	2.2	0.14	5	1	1.2	1.3	0.98	1.2	1.1	1.5	0.38	n/a	n/a	1.7	1.8	2.1	2.1	2.2
	Ad Analytics (AD)	1.8	2.5	2.2	2.2	1.8	8.4	1.2	3.8	n/a	n/a	0.97	0.94	1.5	0.12	0.48	0.99	0.96	0.19	n/a	67	1.1	2	0.54	1.2	1.2
	Click Analytics (CA)	1	1.2	1.5	1.4	1.2	19	27	13	n/a	n/a	0.99	1.1	1.1	0.49	0.56	1	0.3	0.25	n/a	326	1	n/a	1.1	1.1	1
	Linear Road (LR)	1.5	1.4	0.97	0.77	1.2	n/a	n/a	n/a	n/a	n/a	1	0.99	1.6	1.3	1.5	1.1	1.1	1.7	n/a	n/a	0.99	0.83	1.3	1.3	1.2
State+UDF	Log Analyzer (LA)	1	1.2	1.6	1.9	1.8	412	75	51	4.1	4.1	1	1	1.1	0.88	1.3	1	1	0.96	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Bargain Index (BI)	n/a	n/a	n/a	0.79	0.79	n/a	n/a	n/a	n/a	n/a	0.94	0.96	1.1	0.79	0.8	1.1	0.96	2.6	186	107	0.82	0.96	1.1	1.3	0.77
	Smart Grid (SG)	1.6	1.1	0.99	0.56	2	0.74	1	1	0.033	0.027	1	1	1.1	0.052	0.5	91	576	1068	n/a	n/a	0.98	0.35	0.24	0.24	0.23
	Google Cloud Monitoring (GCM)	n/a	n/a	n/a	1.8	0.73	n/a	n/a	n/a	n/a	n/a	1	0.95	1	0.19	0.67	1	0.85	0.22	n/a	599	0.98	0.99	0.91	0.89	0.93
	Spike Detection (SD)	3	18	0.42	n/a	n/a	0.68	0.028	1.1	n/a	n/a	0.96	1	1	0.52	0.8	1.1	0.98	0.93	24	89	0.98	1.1	1.1	1.1	1.1
Stateless	TPC-H (TPCH)	1	1	1.1	1.1	0.92	1	2.3	2.1	1.6	0.89	0.99	0.99	1.2	1.3	1.4	0.98	1	0.99	1.3	1.3	0.97	0.9	1.6	1.6	1.5
	Word Count (WC)	1	1	1	0.82	0.83	1.1	1.1	1.2	n/a	n/a	1	1.1	1.2	1	1.2	0.99	0.85	0.11	1.2	1.2	1	2.2	0.99	0.85	1

(a) Throughput [×] (green = increases)

		Flink					Storm					Spark					Kafka S.					RisingW.				
		L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh
CPU	Sentiment Analysis (SA)	0.9	0.98	1.1	0.53	3	n/a	n/a	n/a	1.9	2.2	0.5	2	1.3	1.7	5.7	0.38	0.13	0.006	n/a	n/a	1.4	1.8	1.4	1.5	1.5
	Traffic Monitoring (TM)	0.074	0.39	0.46	0.18	2.6	0.42	0.42	0.52	0.53	1.1	0.21	9.4	14	2.6	3.3	1.9	0.001	0.002	n/a	2.1	1	1	1	1	1
	Trending Topics (TT)	1.6	1.3	1	1.3	1.3	0.62	0.64	0.38	1.3	2	0.54	1.4	1.7	0.001	0.2	0.61	0.27	0.25	20	23	0.54	0.54	2.2	2.8	6.3
S+N State	Machine Outlier (MO)	1.4	0.53	0.42	10	23	0.43	0.77	0.91	0.89	1.8	0.85	0.59	1.3	1.1	0.76	0.53	0.013	0.028	n/a	n/a	0.13	1.5	1	0.99	5.1
	Ad Analytics (AD)	1.4	7.7	8	178	26	1.2	0.72	2.3	n/a	n/a	0.89	0.43	0.15	0.18	0.84	0.63	0.55	0.16	n/a	9.2	0.5	1	1	1.2	1.2
	Click Analytics (CA)	1.8	9.6	7.7	720	60	0.65	0.98	0.88	n/a	n/a	1.1	1.1	0.89	0.87	0.99	0.86	0.33	0.25	n/a	5.6	0.66	n/a	0.96	0.67	1.5
	Linear Road (LR)	1.1	1.2	0.95	0.7	1.2	0.052	0.016	0.11	6	4.3	0.4	0.28	0.11	0.79	0.61	n/a	3.4	n/a	n/a	n/a	1.2	1.1	2.1	1.1	1.1
State+UDF	Log Analyzer (LA)	1.7	2	14	703	118	0.16	0.92	1	1.3	1.5	0.83	0.93	1.2	0.76	1.2	1.1	0.92	1	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Bargain Index (BI)	1.1	0.83	3.3	31	4.8	1.7	3.3	2.9	n/a	n/a	1	0.91	1.3	0.88	0.93	0.1	0.86	0.89	6.5	6.3	2.3	1.2	0.7	0.7	0.95
	Smart Grid (SG)	1.1	1.6	1.8	2.4	17	0.85	0.21	0.53	1.9	1.5	0.99	0.79	0.69	0.12	0.48	1.8	1.3	1.1	n/a	n/a	1	0.69	0.017	0.026	0.026
	Google Cloud Monitoring (GCM)	n/a	n/a	n/a	0.58	1.2	n/a	n/a	n/a	n/a	n/a	0.91	0.81	0.77	0.14	0.52	1.1	0.76	0.61	n/a	11	1	1	1	1.2	1.2
	Spike Detection (SD)	1.1	0.67	3.3	93	12	0.16	2.2	2.7	n/a	n/a	0.99	0.85	0.87	1	0.51	0.77	0.8	0.3	15	12	0.99	0.69	0.67	0.96	1.5
Stateless	TPC-H (TPCH)	2	2.3	1.4	1.1	1.8	0.65	1.4	1.5	1.3	1.7	1.2	0.87	0.91	0.94	0.97	0.28	0.28	0.12	0.51	4.2	1	1	0.99	1.2	1.2
	Word Count (WC)	1.2	2.8	2.4	2.1	1.2	7.5	7.7	7.6	n/a	n/a	1	1.1	2.3	3.1	1.5	0.34	0.043	0.002	4.4	4.4	2.2	1	2.2	2.2	2.2

(b) E2E latency [×] (green = decreases)

		Flink					Storm					Spark					Kafka S.					RisingW.				
		L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh	L	M	H	DS2	Dh
CPU	Sentiment Analysis (SA)	+49	+51	+48	+45	-1	+56	+52	+52	+14	+8	+0	+0	-0	-1	-1	+4	+10	+11	-30	-31	+1	+2	+1	+2	+11
	Traffic Monitoring (TM)	+20	+55	+55	+52	+2	+57	+55	+53	+49	+16	-0	+0	+0	-1	-1	+11	+62	+16	n/a	+0	+0	-0	+1	+1	+1
	Trending Topics (TT)	+6	+13	+28	+37	+15	+16	+19	+11	+12	-4	+0	+0	-0	-1	-1	+6	+13	+35	+19	+3	+1	+0	-0	+1	+1
S+N State	Machine Outlier (MO)	+6	+13	+21	+7	-4	+25	+39	+37	+27	+9	+0	+0	+0	-1	-1	+3	+22	+62	-28	-28	+1	+3	+1	+3	+0
	Ad Analytics (AD)	+8	+9	+8	-10	-13	+44	+16	+19	+12	+12	-0	+0	+0	-1	-1	+7	+22	+49	n/a	+3	-0	+0	+0	+1	-0
	Click Analytics (CA)	+9	+5	+9	-5	-4	+41	+41	+33	+19	+19	+0	+0	+0	-1	-1	+15	+51	+51	n/a	-17	+0	+1	+0	+1	+1
	Linear Road (LR)	+3	+3	+5	+3	-1	+57	+63	+59	+27	+16	+0	+0	-0	-1	-1	+4	+11	+37	n/a	-25	-0	-1	+1	-0	+1
State+UDF	Log Analyzer (LA)	+5	+13	+14	-5	-8	+16	+25	+33	+7	-9	-0	+0	+0	-1	-1	+4	+11	+17	-25	-25	n/a	n/a	n/a	n/a	n/a
	Bargain Index (BI)	+3	+12	+16	-4	-4	+31	+27	+23	+20	+20	+0	+0	+0	-1	-1	+4	+5	+6	+3	-3	+0	+0	+0	-0	-0
	Smart Grid (SG)	+3	+5	+12	+9	-6	+25	+16	+17	+2	+5	-0	+0	+0	-1	-1	+16	+32	+46	-20	-20	+1	+1	+1	+1	+1
	Google Cloud Monitoring (GCM)	n/a	n/a	n/a	+6	-1	+49	+53	+49	+39	+39	+0	-0	-0	-1	-1	+9	+11	+15	n/a	-6	-0	+1	+1	-0	+0
	Spike Detection (SD)	+3	+11	+20	-3	-2	+39	+20	+40	+9	+10	+0	-0	-0	-1	-1	+4	+7	+11	+38	-2	+1	+1	+1	+1	+1
Stateless	TPC-H (TPCH)	+2	+6	+11	+7	+2	+26	+18	+14	+14	+4	+0	-0	+0	-1	-1	+3	+5	+7	-0	-2	+0	+0	+0	+0	+1
	Word Count (WC)	+3	+1	-3	+2	+1	+24	+22	+18	+36	+36	-0	-0	+0	-1	-1	+10	+38	+50	-0	-2	+2	+6	+0	-1	-0

(c) CPU utilization [pp] (%-points)